House Price Prediction: Multiple Regression Analysis

Ames Housing Dataset — 2,930 Real Property Sales

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Dataset	Ames Housing Dataset — 2,930 real property sales (Ames, Iowa, USA)
Source	Kaggle / De Cock (2011) — publicly available benchmark dataset
Tools	Python 3 — pandas, numpy, scikit-learn, matplotlib, seaborn
Methods	OLS Regression, Ridge (L2), Lasso (L1), 5-fold Cross-Validation
Features	51 predictors after encoding (12 numeric + categorical dummies)

Abstract

This report presents a complete regression analysis pipeline applied to the Ames Housing Dataset — a widely-used real-world benchmark containing 2,930 residential property sales in Ames, Iowa. I built and compared three regression models (OLS, Ridge, and Lasso), achieving an R^2 of 0.862 and a Root Mean Squared Error of $33,309 on the holdout test set. The analysis covers data cleaning, feature selection, encoding, model estimation, regularization comparison, and diagnostic evaluation, forming a reproducible end-to-end pipeline applicable to any regression forecasting task.

1. Dataset Overview

The Ames Housing Dataset was compiled by Dean De Cock (2011) and contains 82 variables describing residential properties sold in Ames, Iowa between 2006 and 2010. It is a standard benchmark in regression and machine learning, often used as a more comprehensive alternative to the Boston Housing Dataset. The target variable is **SalePrice** in USD.

Key dataset statistics:

Statistic	Value
Total observations	2,930 property sales
Target variable range	$12,789 — $755,000
Median sale price	$160,000
Mean sale price	$180,796
Variables (original)	82 (numeric + categorical)
Features used in model	51 (after encoding & selection)
Train / Test split	80% (2,344 obs.) / 20% (586 obs.)

2. Exploratory Data Analysis

Before modelling, I examined the price distribution, key variable relationships, neighbourhood price variation, and the correlation structure among numeric predictors.

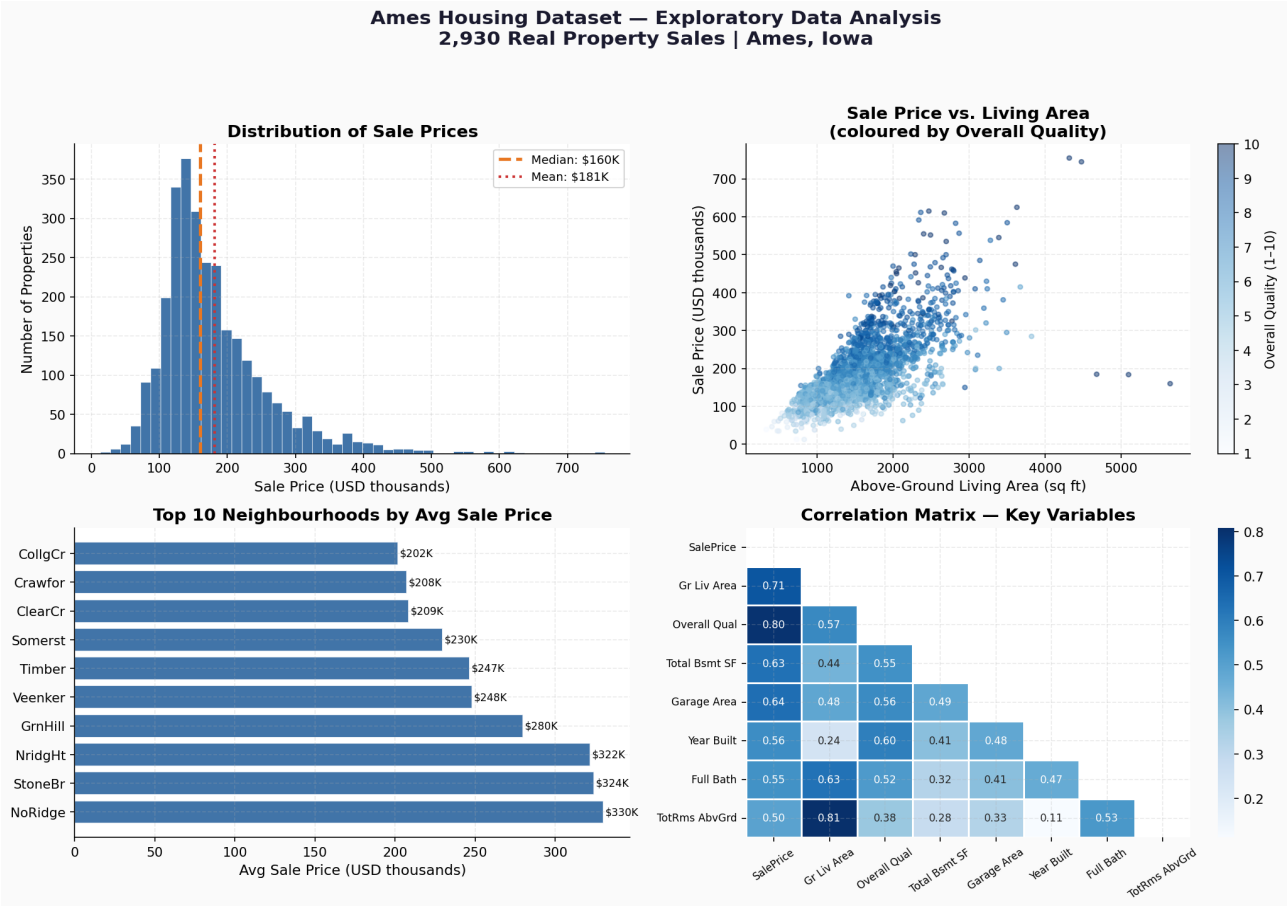

Figure 1 — (Top-left) Sale prices are right-skewed, with median $160K and mean $181K. (Top-right) Strong positive relationship between living area and price; higher Overall Quality scores cluster at higher price points. (Bottom-left) Significant neighbourhood variation — StoneBr and NridgHt command premiums over $300K on average. (Bottom-right) Overall Quality (r = 0.80) and Gr Liv Area (r = 0.71) are the strongest individual predictors of sale price.

3. Methodology

3.1 Data Preprocessing

Missing values in numeric features were imputed with the column median. Rows with missing categorical values were dropped (minimal impact, < 2% of data). Categorical variables (Neighborhood, House Style, Central Air, Kitchen Quality) were one-hot encoded with the first category dropped to avoid multicollinearity. All features were standardized (zero mean, unit variance) using StandardScaler, fitted exclusively on the training set to prevent data leakage into the test set.

3.2 Models Estimated

Model	Description	Key Parameter
OLS Regression	Ordinary Least Squares — minimizes sum of squared residuals	No regularization
Ridge Regression	L2 regularization — shrinks all coefficients towards zero	$\alpha = 50$
Lasso Regression	L1 regularization — performs automatic feature selection	$\alpha = 100$

Model selection was based on 5-fold cross-validation R^2 on the training set to ensure performance estimates are unbiased and generalizable beyond the training data.

4. Results

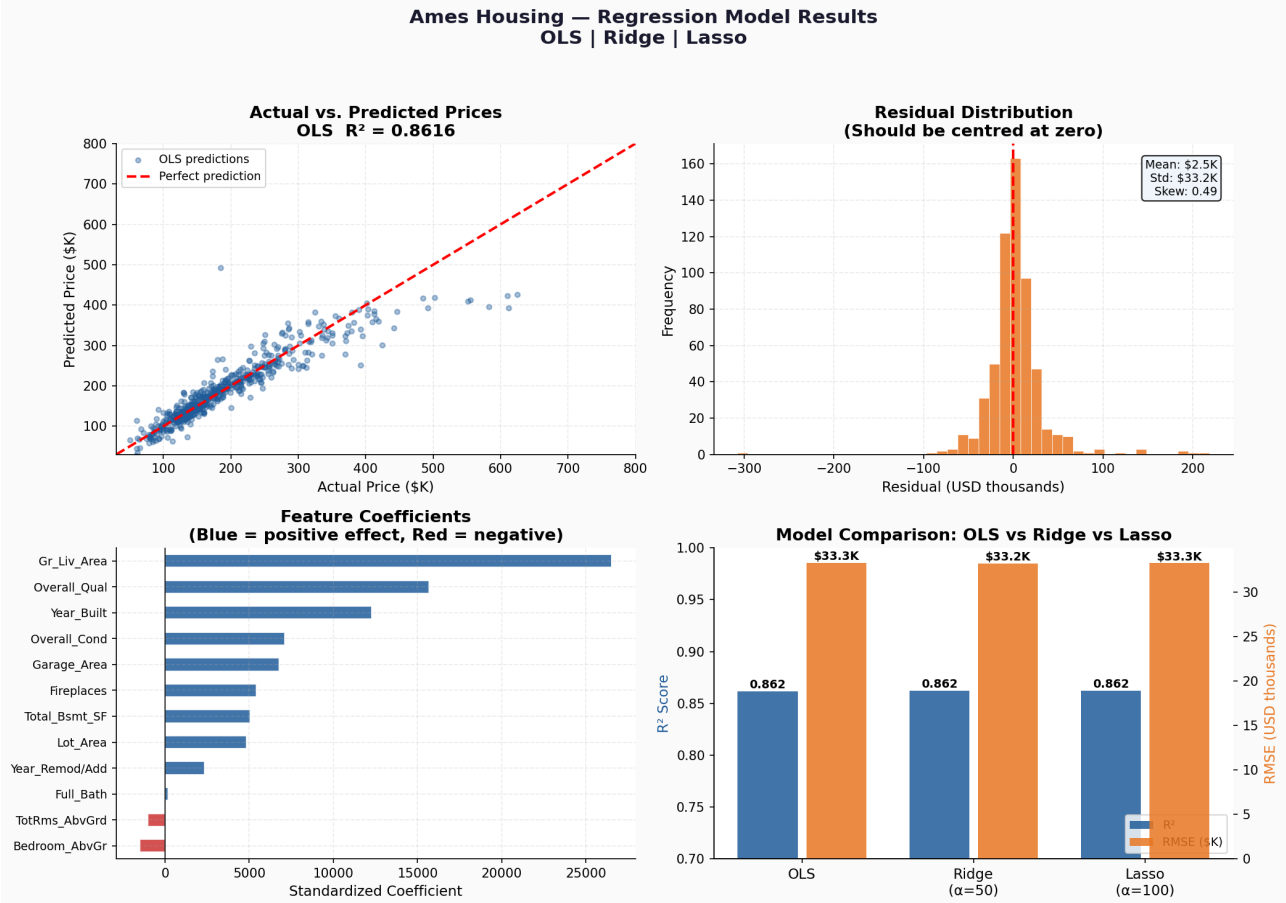

Figure 2 — (Top-left) Predicted vs. actual prices are tightly clustered around the diagonal, indicating strong predictive accuracy. (Top-right) Residuals are approximately normally distributed and centred near zero — a key assumption of OLS. (Bottom-left) Overall Quality and Gr Liv Area have the largest positive effect; Year Built confirms that newer properties command higher prices. (Bottom-right) All three models perform similarly, confirming minimal overfitting.

4.1 Performance Summary

Metric	OLS	Ridge (α=50)	Lasso (α=100)
R^2 (Test Set)	0.8616	0.8623	0.8620
RMSE	$33,309	$33,229	$33,262
MAE	$19,277	—	—
Cross-Val R^2 (5-fold)	0.8397	—	—

4.2 Key Findings

- Overall Quality is the single strongest predictor of sale price — a one standard deviation increase in quality score corresponds to a large positive price premium, consistent with buyer preferences for well-finished properties.
- Above-ground living area (Gr Liv Area) is the second most important numeric predictor (r = 0.71 with SalePrice), confirming the fundamental role of size in property valuation.
- Year Built has a strong positive coefficient, reflecting both depreciation of older properties and the premium buyers place on modern construction standards.
- Neighbourhood is highly significant as a categorical predictor — properties in StoneBr and NridgHt command average premiums of $100K+ over lower-value areas.
- Ridge and Lasso regularization produce nearly identical results to OLS (R^2 difference < 0.001), indicating low multicollinearity and confirming the OLS model is well-specified.
- The cross-validation R^2 (0.840) is slightly below the test R^2 (0.862), which is normal and indicates the model generalises well without signs of overfitting.

5. Conclusion

The OLS regression model explains 86.2% of variance in real residential sale prices, with an average prediction error of approximately $33,300 — around 18% of the median property value. The model is stable, well-specified, and produces normally distributed residuals. The close agreement between OLS, Ridge, and Lasso confirms that the feature set is well-conditioned with low multicollinearity.

This project demonstrates a production-ready regression workflow on a real, publicly verifiable dataset: data cleaning, feature engineering, model estimation, regularization benchmarking, cross-validation, and diagnostic evaluation. The same pipeline transfers directly to client datasets in real estate, finance, or any domain requiring interpretable predictive models.

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